Liquid intrusion and evaporation in low-pressure cavities: Numerical investigation with STAR-CCM+

Harsh Pandey

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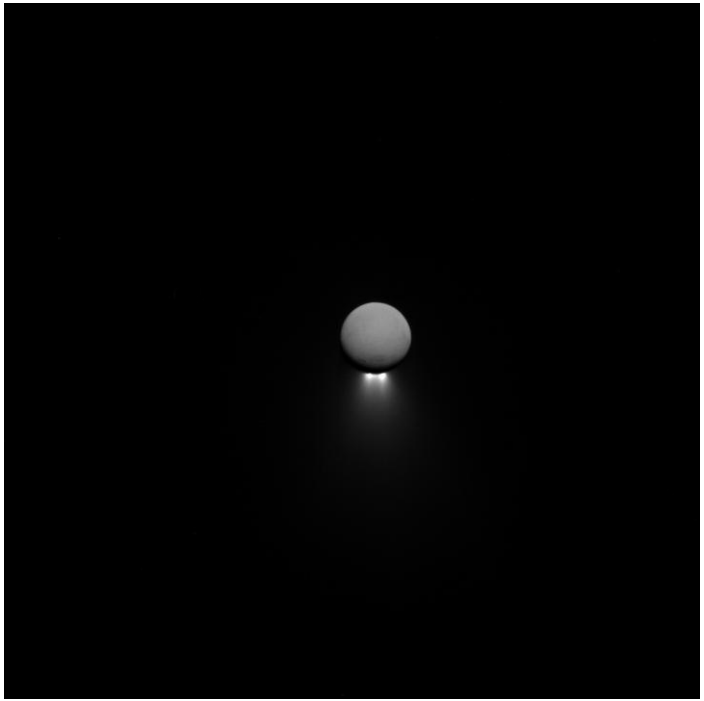
Cryovolcanism on Icy Moons



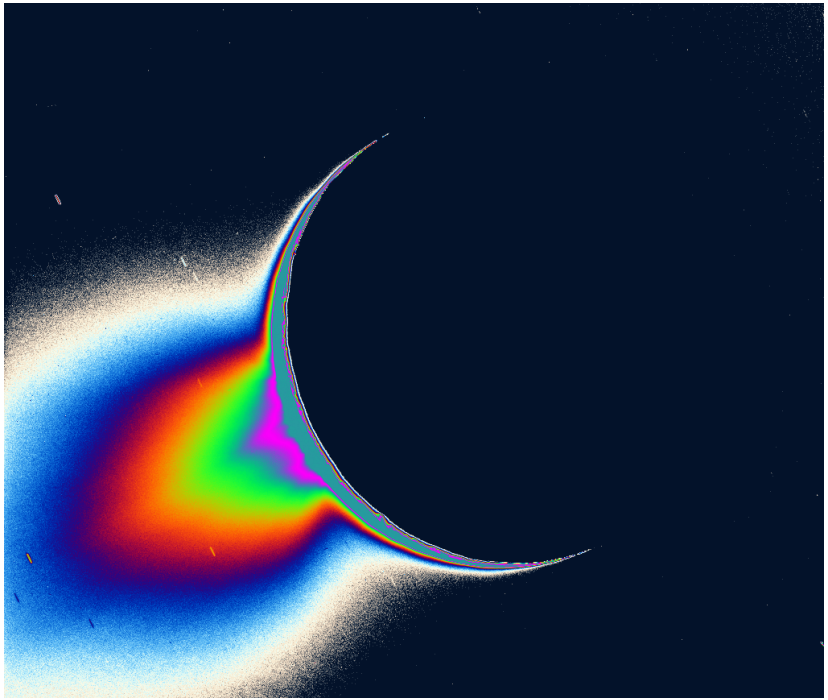
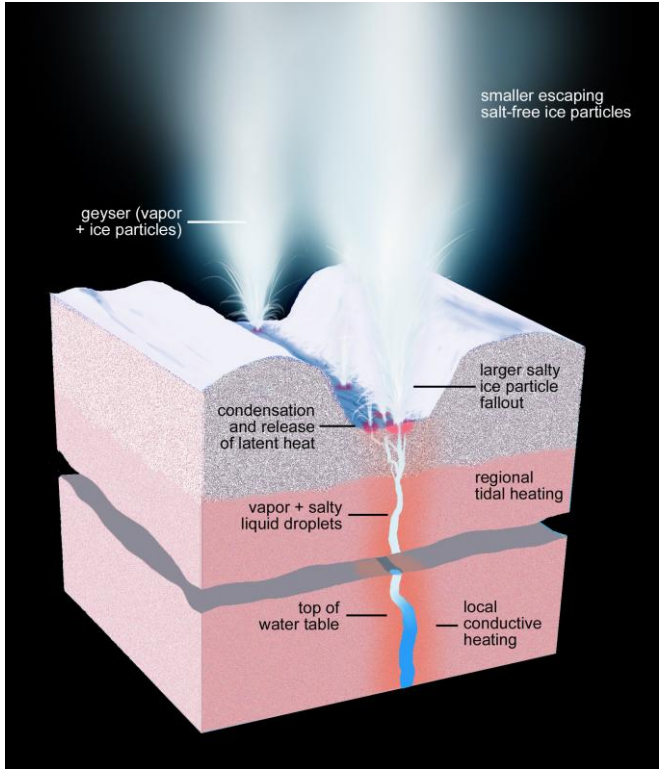


Image Credits: NASA/JPL/Space Science Institute

The Flash Evaporation Experiment



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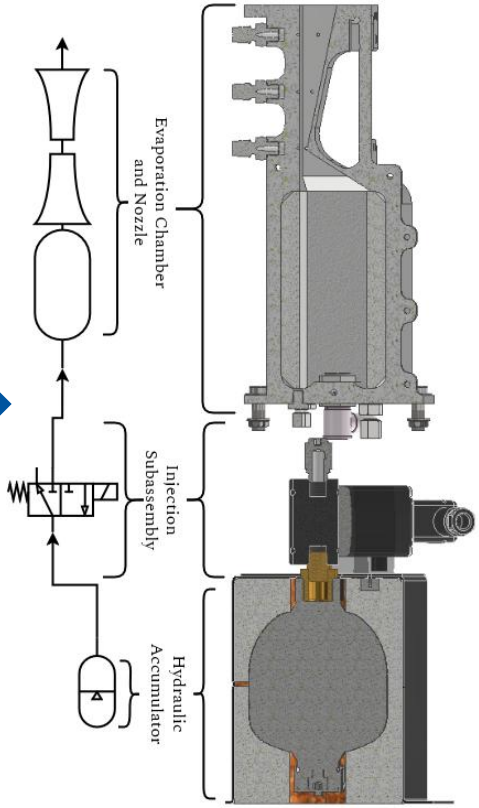

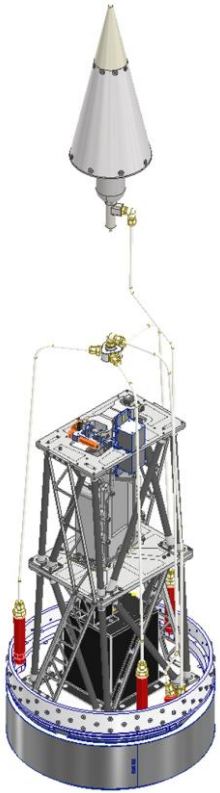

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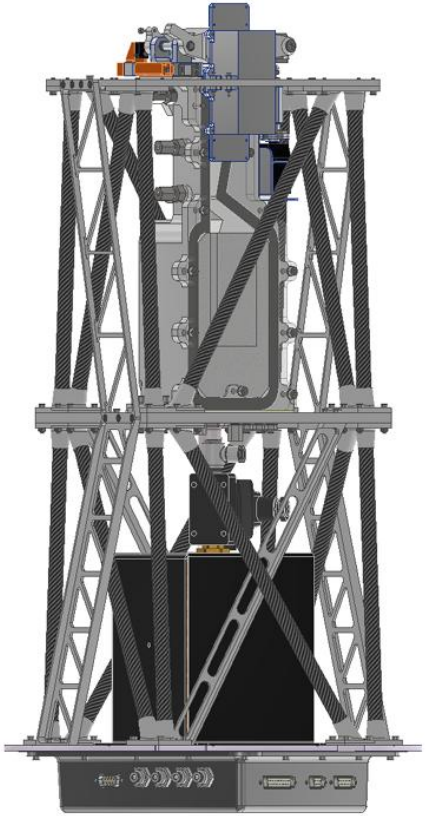


Outline

Subproblem 1:
Liquid infiltration

Subproblem 2:
Flash evaporation

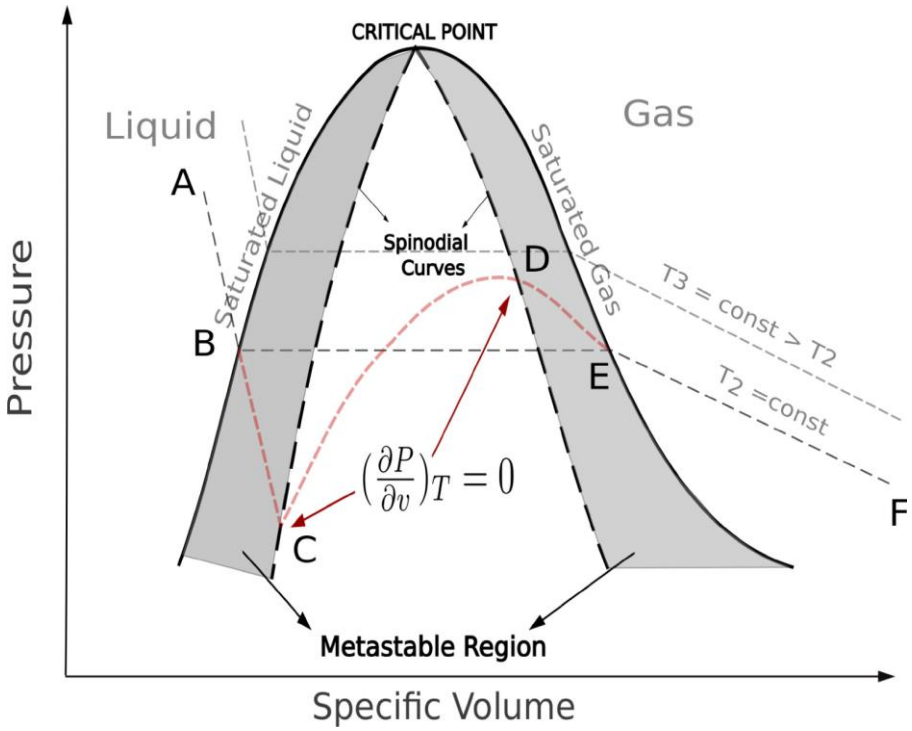
1. Flash evaporation – a review
2. Modelling liquid-gas two-phase flows
   - Compare formulations
   - Compare STAR-CCM+ implementations
3. Modelling the liquid infiltration of the evaporation chamber

Flash Evaporation

1. Metastability
2. Stages of flash evaporation
3. Modelling considerations

Flash Evaporation and Metastability

- Intense vaporisation triggered by depressurization of superheated fluids below saturation pressure
- No external heat added to system
- Vaporisation observed in the liquid bulk and on the free-surface



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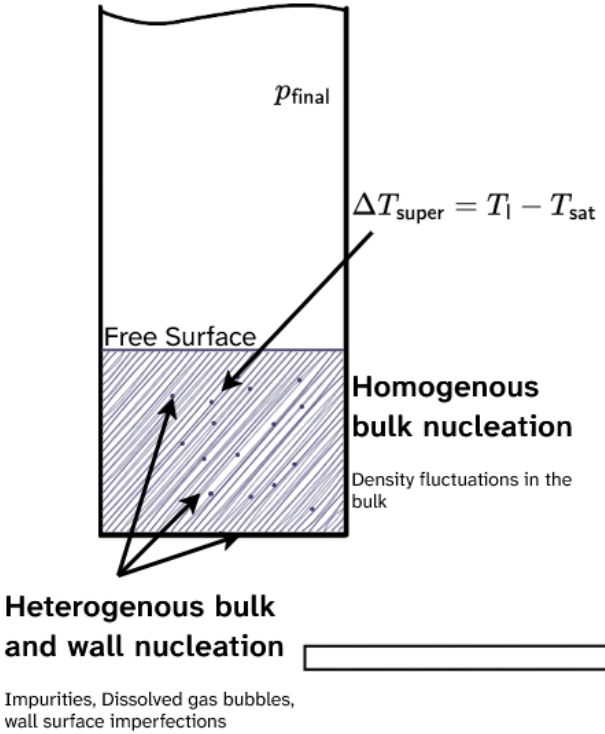
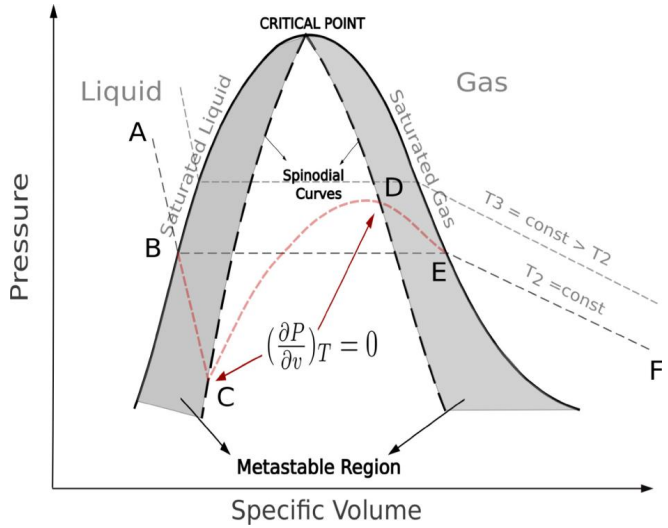
Stages of the Flash Evaporation Phenomenon





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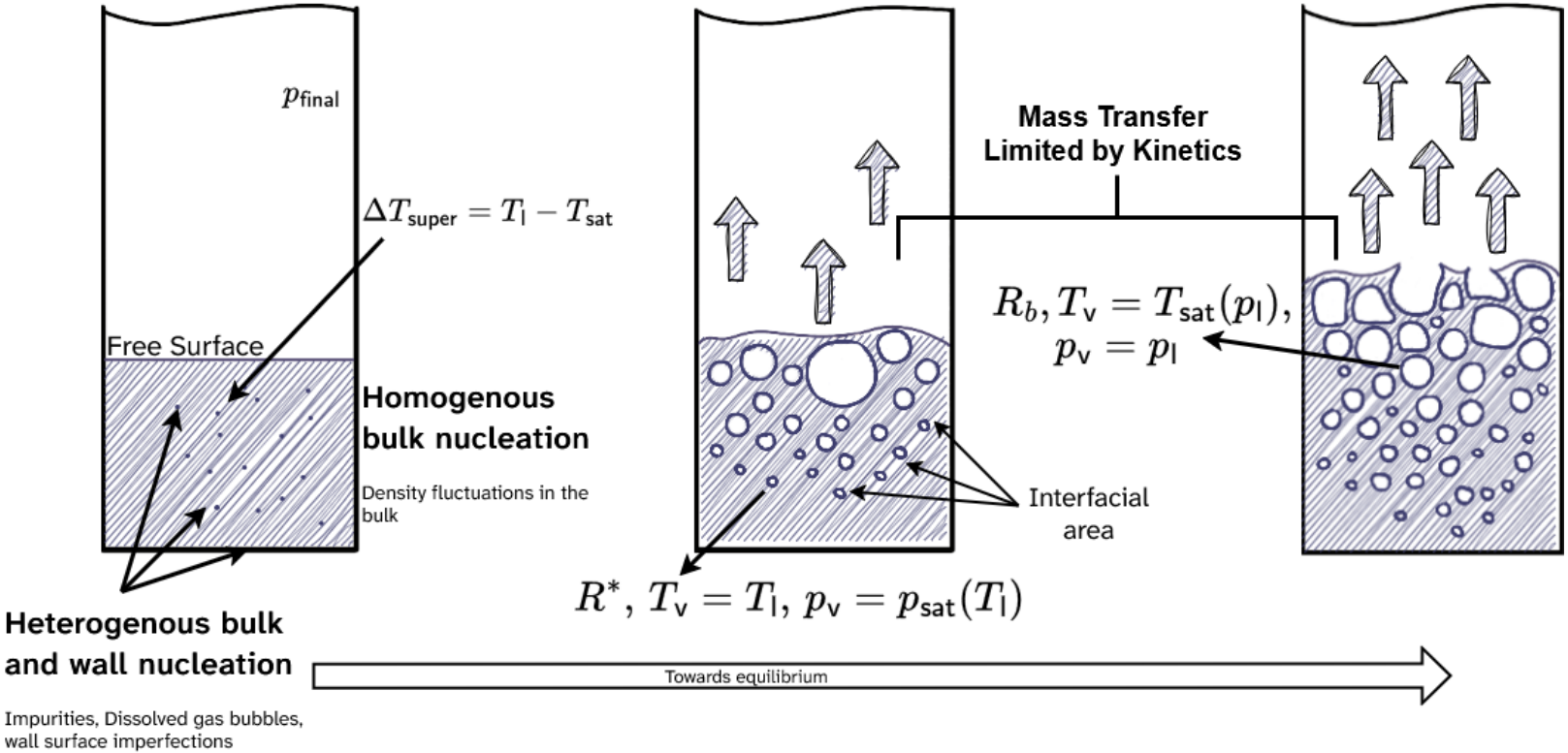
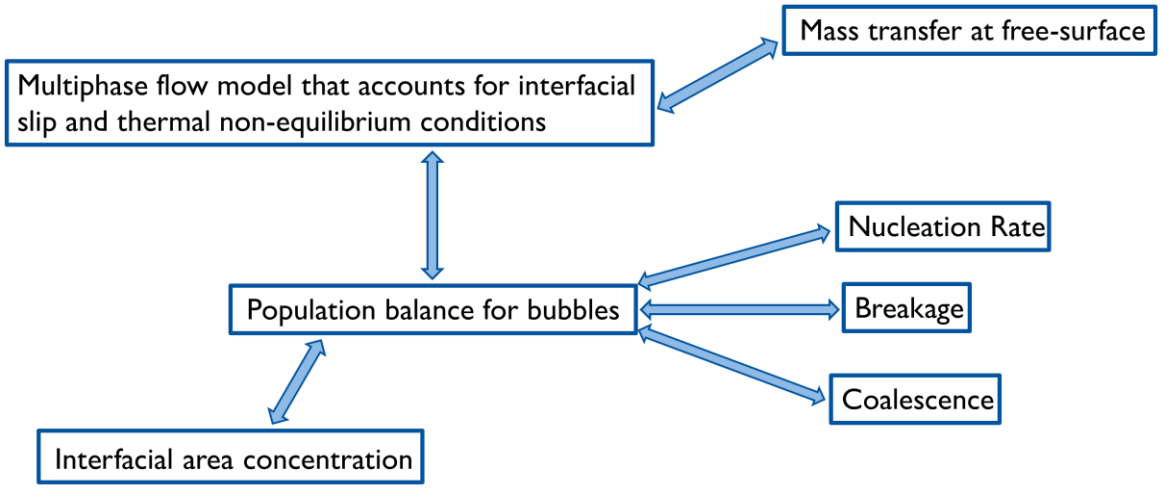
Towards equilibrium

Stages of the Flash Evaporation Phenomenon



Implementation and Limitations



- Pressure-driven mass transfer inaccurately considered
- Inaccurate estimation of interfacial area concentration
- Nucleation is stochastic
- Uncertain closures
- Environmental Sensitivity

Liquid-Gas Two-Phase Flows

1. Multiphase Flow Formulations
2. STAR-CCM+ Implementation
3. Studying the Rayleigh-Taylor Instability
4. Adaptive Mesh Refinement for Free-Surface Flows

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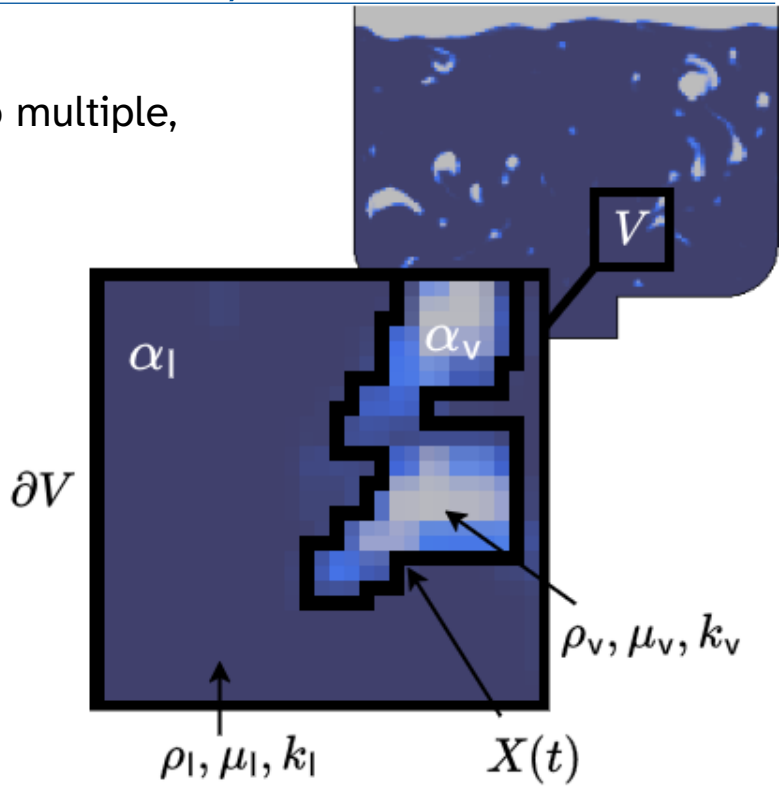
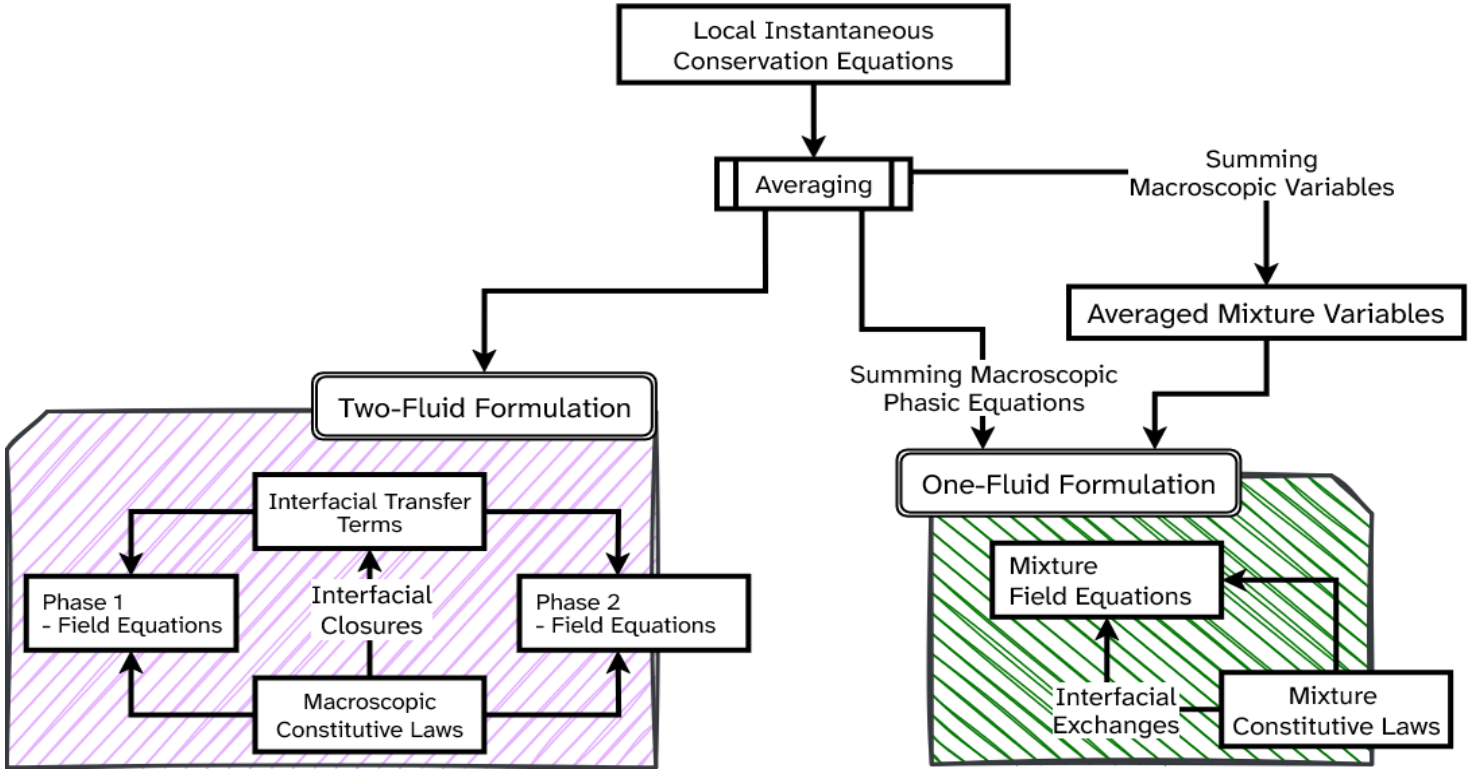
Dealing with Scales (and Computational Limits)

Direct simulation is computationally impractical due to multiple, deformable moving interfaces.

$$\alpha_i(\mathbf{x}, t) = \frac{V_i}{V}, \qquad \alpha_l + \alpha_v = 1$$



Two Fluids or a Mixture



Comparing the Formulations

	Two-Fluid Formulation	One-Fluid Formulation
Approach	- Set of governing equations **for each phase** - Phases treated as **interpenetrating continua** - they can coexist everywhere in the flow domain	- **Single set** of governing equations - Treats all phases as a **single fluid with variable properties**
Pros	- More suitable to **disperse flows** - can account for rapid **non-equilibrium phenomenon**	- Capturing **complex interface topologies** - separated, **free-surface flow**
Cons	- Requires **extensive closure relations**, which can be difficult to validate. - Can suffer from **numerical instabilities** or "ill-posedness" in certain flow conditions.	Not ideal for modeling flows with **significant mass transfer** or **strong thermal non-equilibrium** between phases
STAR-CCM+ Implementation	**Eulerian Multiphase (EMP) Model**	**Volume-of-Fluid (VOF) Method**

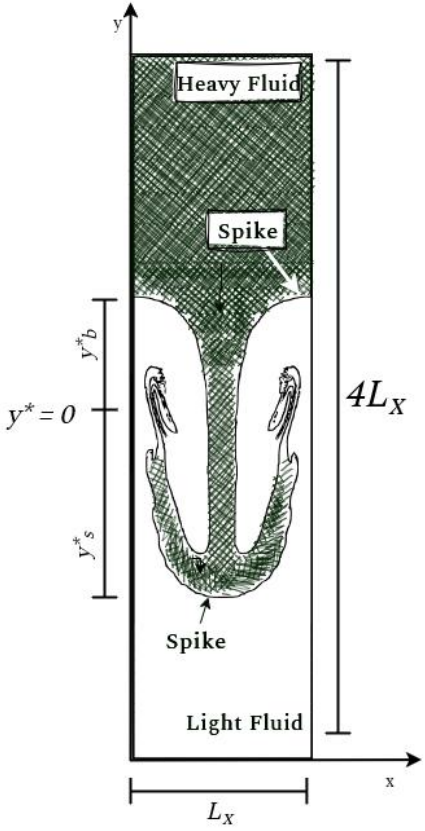
Rayleigh-Taylor Instability

Between two fluids when a denser fluid is accelerated into a lighter one

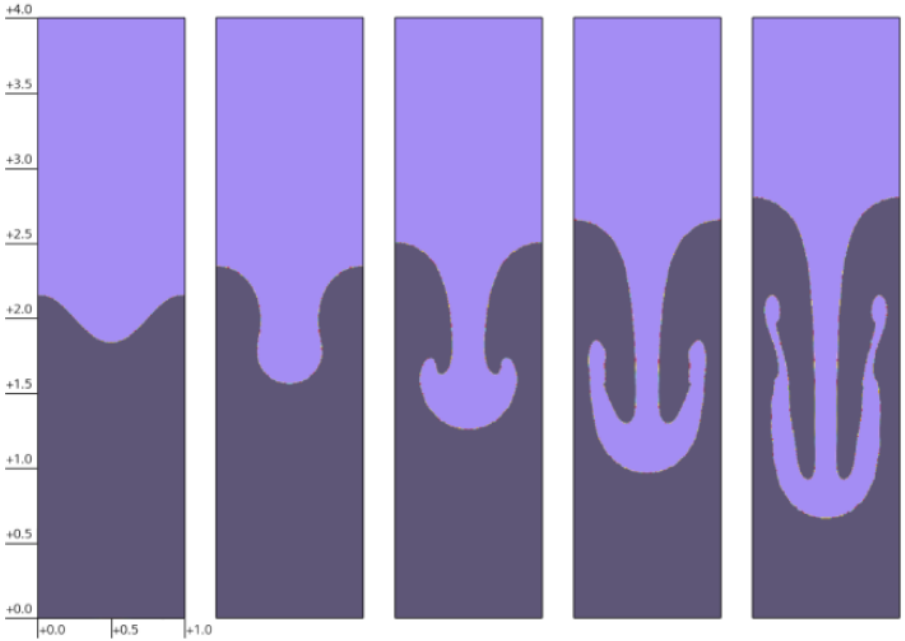
Why?
It is a classical benchmark for validating two-phase flow solvers

- Complex evolution of the moving interface
- Instability development highly sensitive to interface dynamics

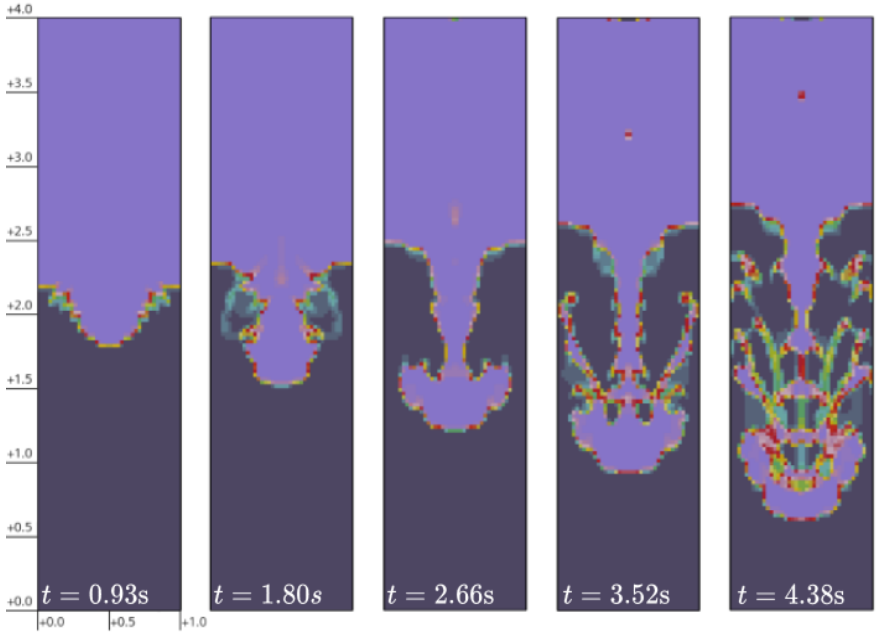
Key Parameters:
- Reynolds Number
- Absolute distance to spike and bubble tips as measured from the centre of the domain, marked by y*



VOF vs EMP: RT Instability

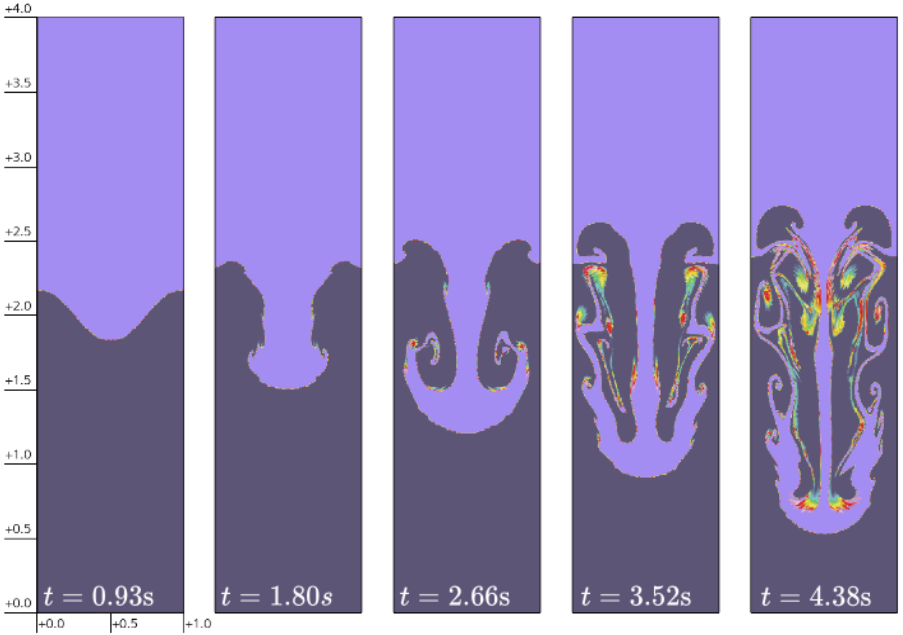


VOF: Reynolds Number = 300

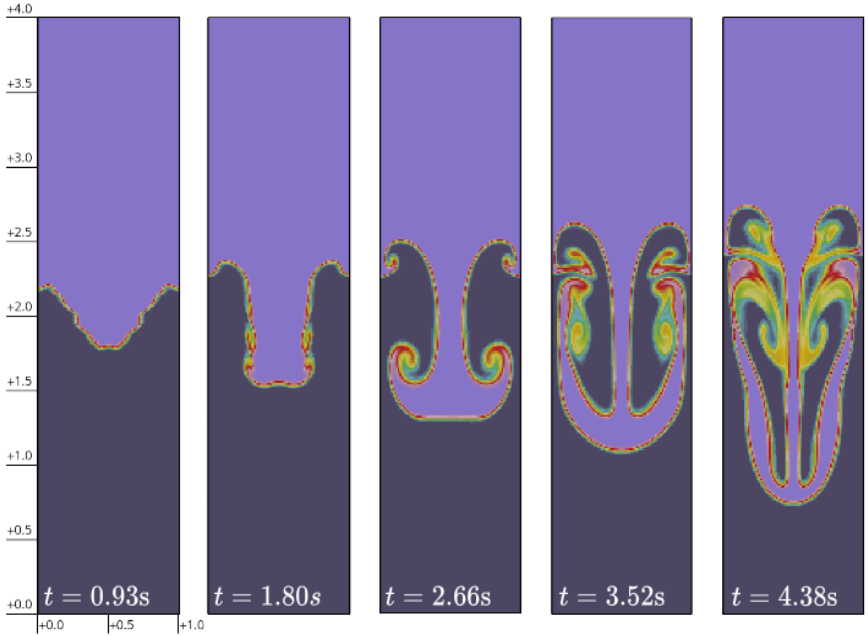


EMP: Reynolds Number = 300

VOF vs EMP: RT Instability

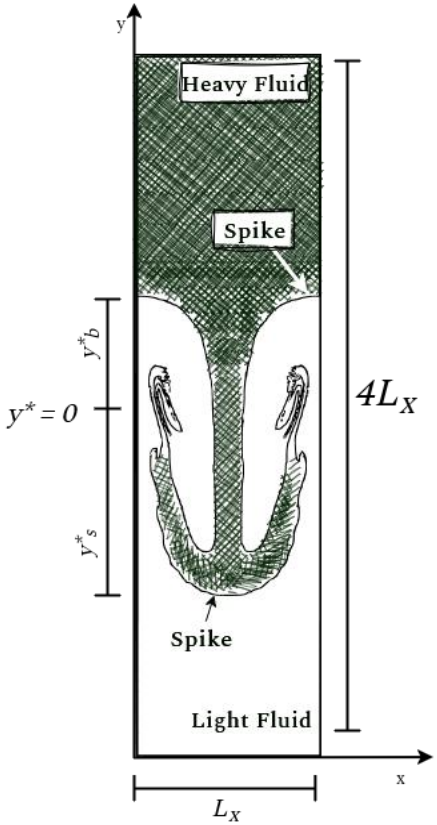


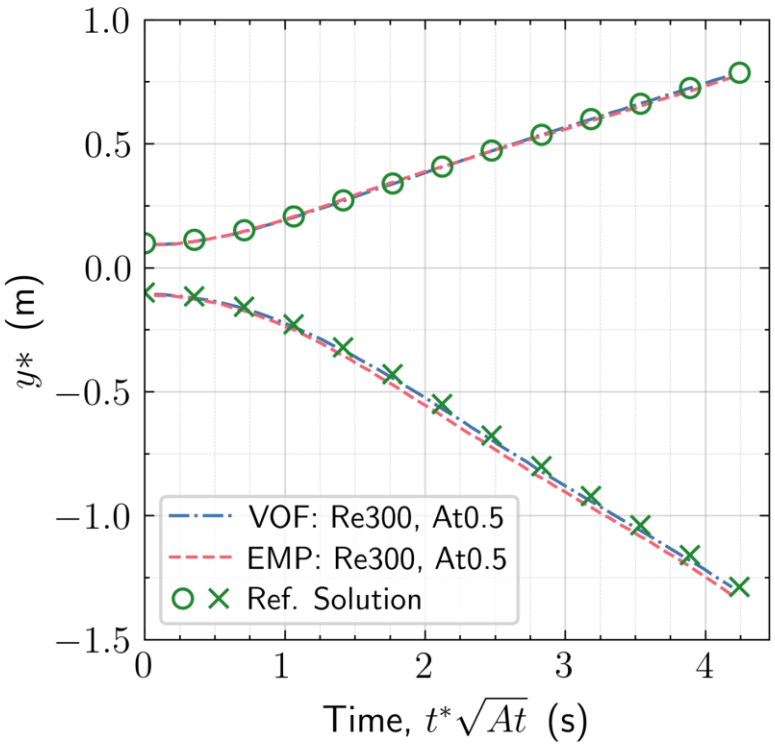
VOF: Reynolds Number = 3000



EMP: Reynolds Number = 3000

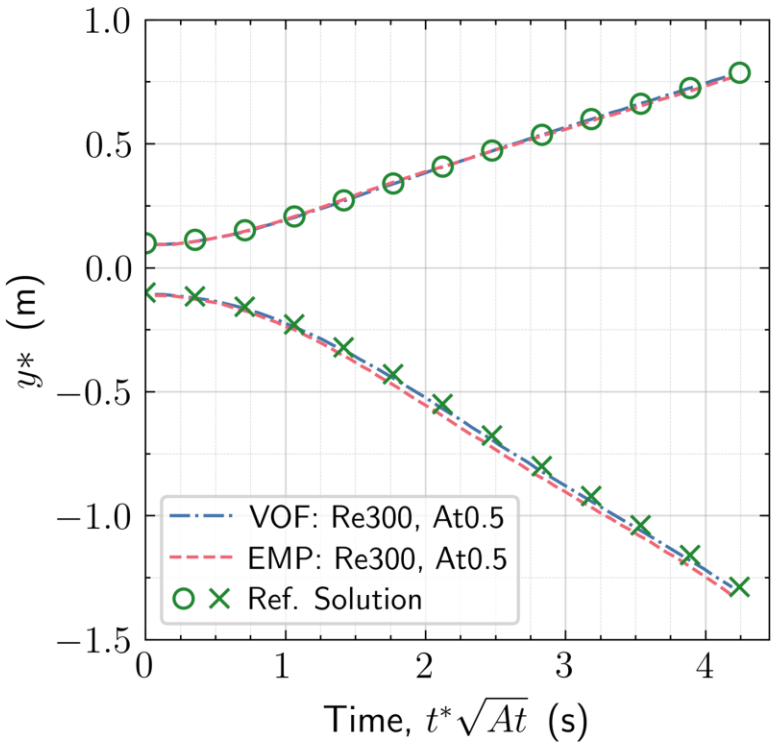
VOF vs EMP: Bubble and Spike Evolution



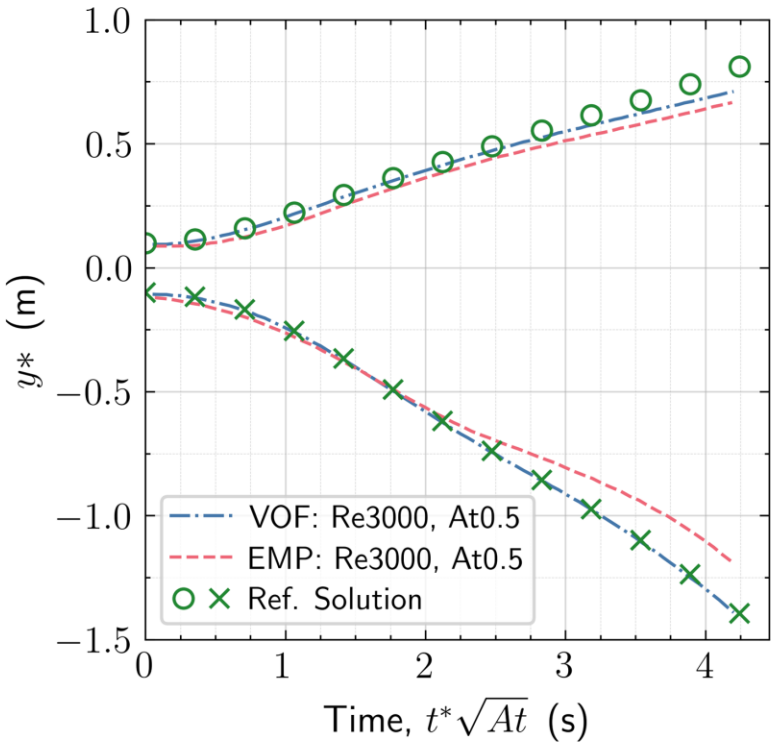


Comparison – VOF and EMP: Reynolds Number = 300

VOF vs EMP: Bubble and Spike Evolution



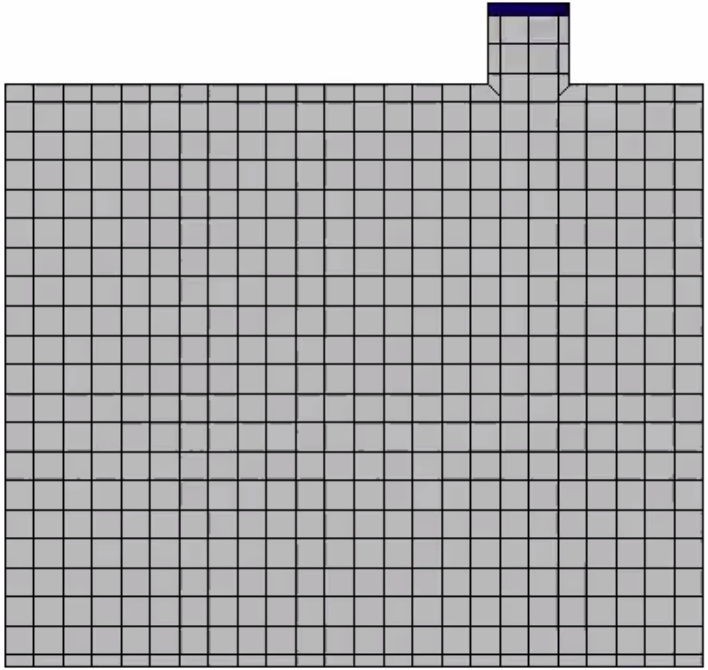
Comparison – VOF and EMP: Reynolds Number = 300



Comparison – VOF and EMP: Reynolds Number = 3000

Adaptive Mesh Refinement

- Automated AMR unavailable for 2D meshes
- A general Remeshing model that triggers remeshing when certain conditions are met
- Mesh refinement unidirectional



Summary

Even with Large Scale Interface detection, VOF objectively better at handling free surface flows

Infiltration Subproblem	**Evaporation Subproblem**
Large, distinct interface	Dispersed and transitional flow
Interphase slip or non-equilibrium assumptions unnecessary	Non-equilibrium effects dominant
Volume-of-Fluid Method	**(proposed) Eulerian Multiphase Model**

Liquid Infiltration of the Evaporation Chamber

1. Computational Setup
2. Evolution of the Interface and Flow Field

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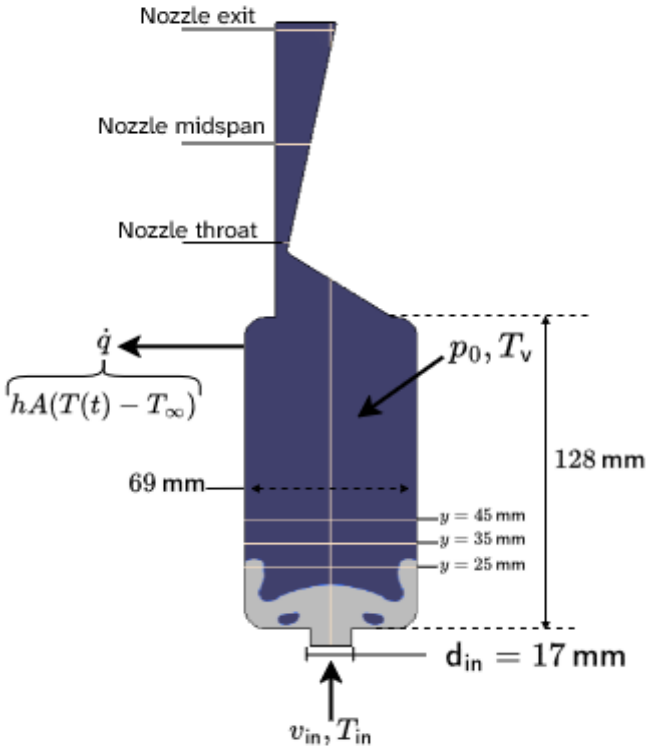
Modelling Infiltration

Ensuring Similarity

- Evaporation Chamber Dimensions
  - Length x Width x Height = 128 mm x 69 mm x 50 mm
- Computational Domain Dimensions
  - Length x Width x Height = 128 mm x 69 mm x 1000 mm
- Flow rate: 15mL/s, Discharge time: 10s
- Reference length scale: d_{in}
- Reference velocity scale: $v_{in} = \frac{0.00015}{d_{in} t_{dis}}$



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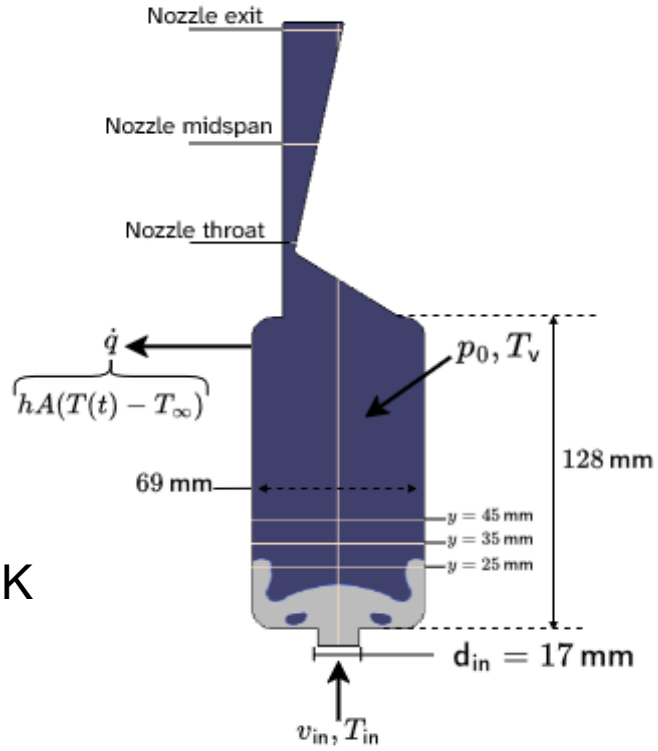
Modelling Infiltration

Boundary Conditions

- Convective heat transfer along walls

$$-\frac{1}{h} = \frac{1}{h_{Al}} + \frac{1}{h_{ins}}$$

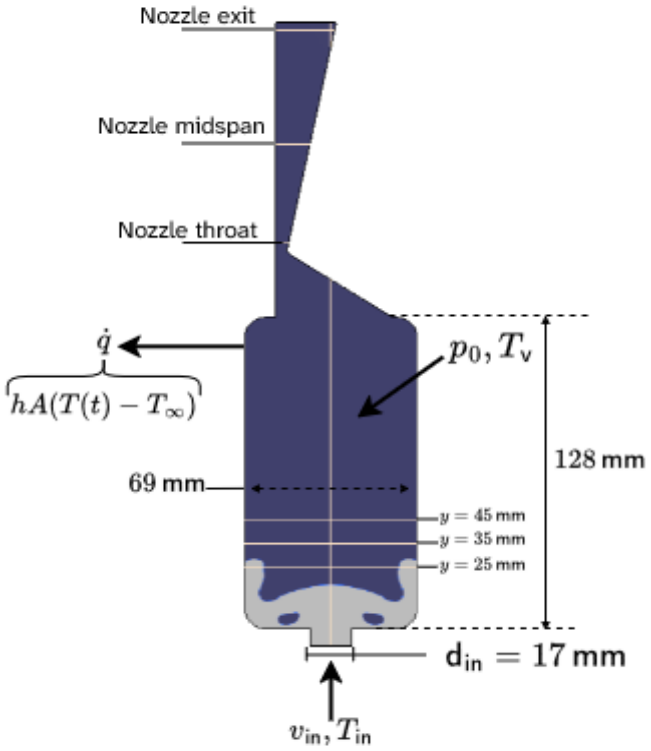
- Adiabatic outlet
- Specified temperature of water at inlet, 353.15K
- Initial temperature inside chamber, 368.15K
- No slip along walls
- Velocity inlet (velocity specified)



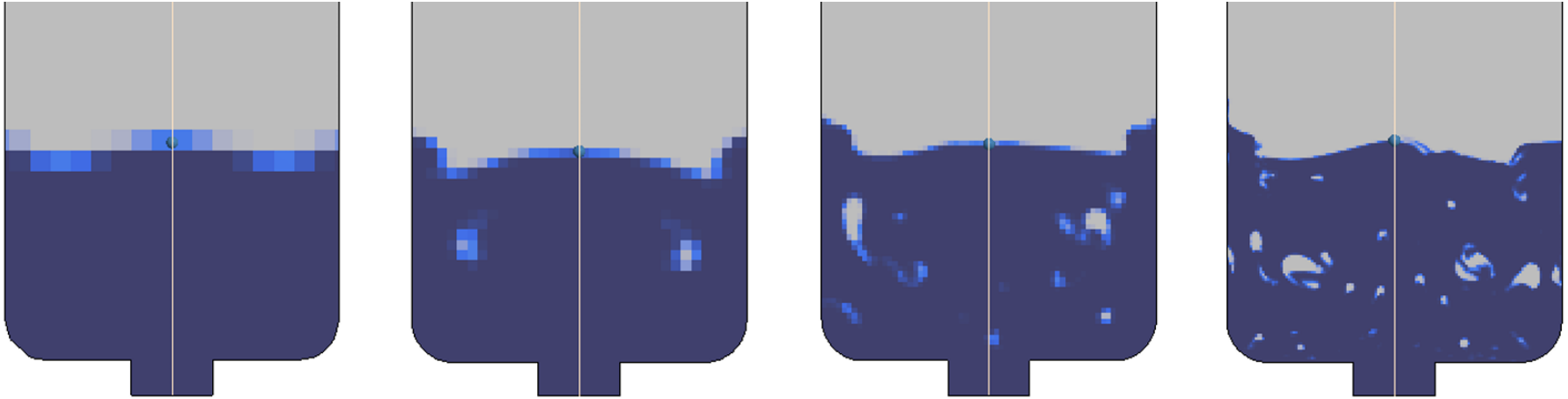
Modelling Infiltration

Measurement Locations

- Section planes in nozzle
- Section planes in evaporation chamber
- Section plane along the centerline

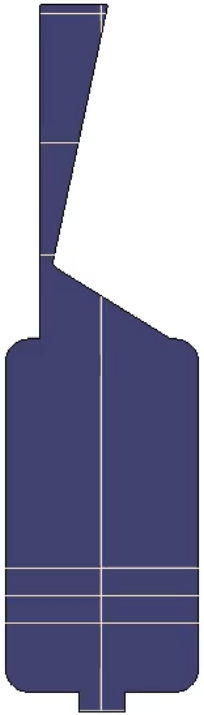


Sharper Features with Increasing Refinement

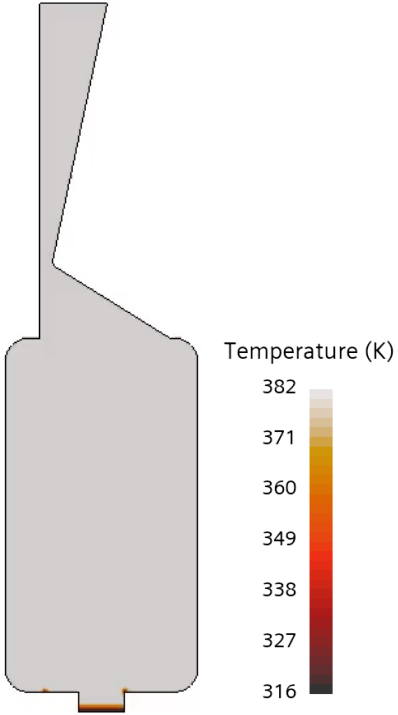


Interface at time, t = 3.25 s, with increasing mesh refinement from left to right - $\Delta h_1 = d_{in}/4$, $\Delta h_2 = d_{in}/8$, $\Delta h_3 = d_{in}/16$, $\Delta h_4 = d_{in}/32$

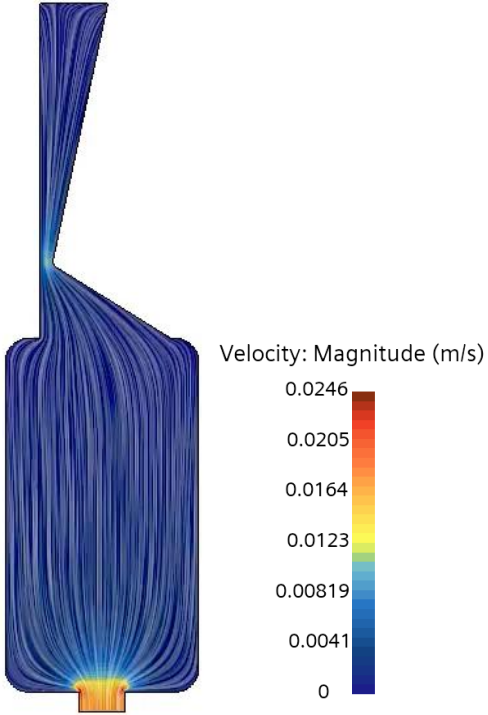
Evolution of the Flow Field



Volume Fraction of Water

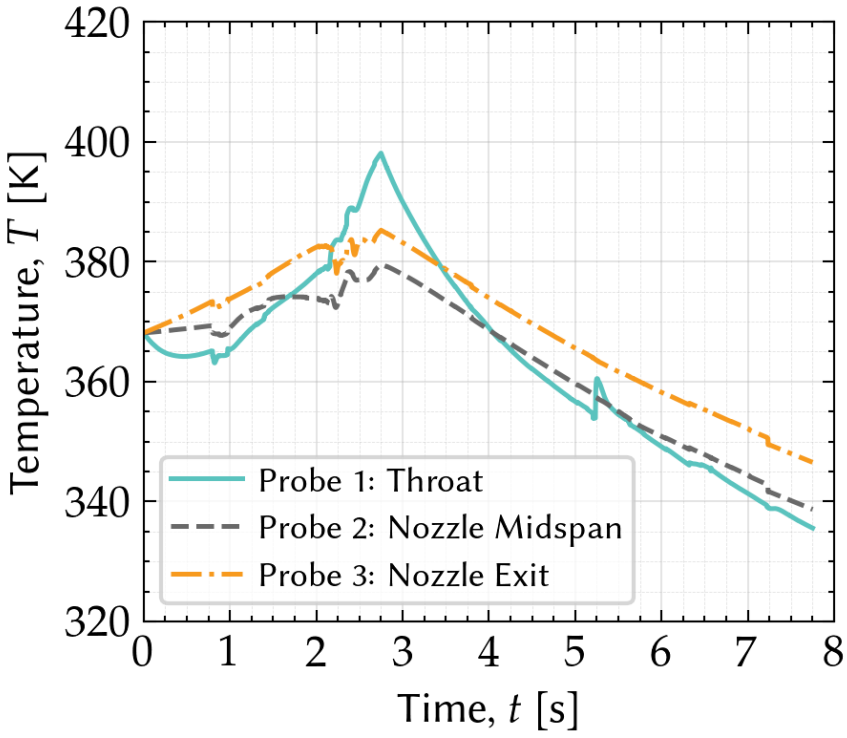


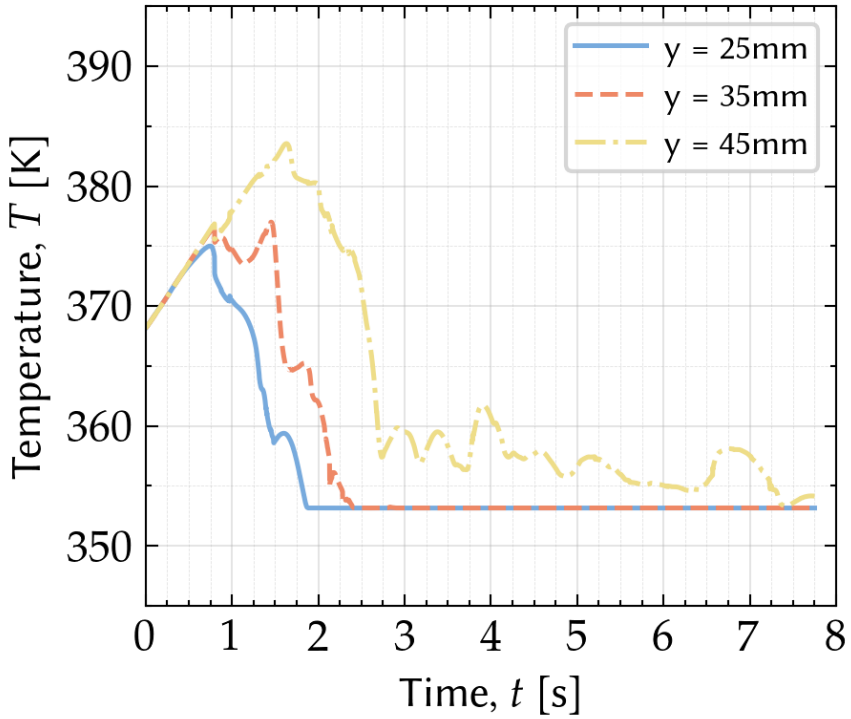
Temperature



Velocity Streamlines

Measured Temperatures





Conclusions

- Identifying what's important to consider in modelling flash evaporation
- Reviewing and comparing the one-fluid and two-fluid models
- Understanding the capabilities and limitations of STAR-CCM+
- Modelling the infiltration subproblem and estimating the flow characteristics at the start of the evaporation problem

Relevant sim files and thesis found at:
https://doi.org/10.5281/zenodo.17156697

Thank you for your attention (:

Questions?